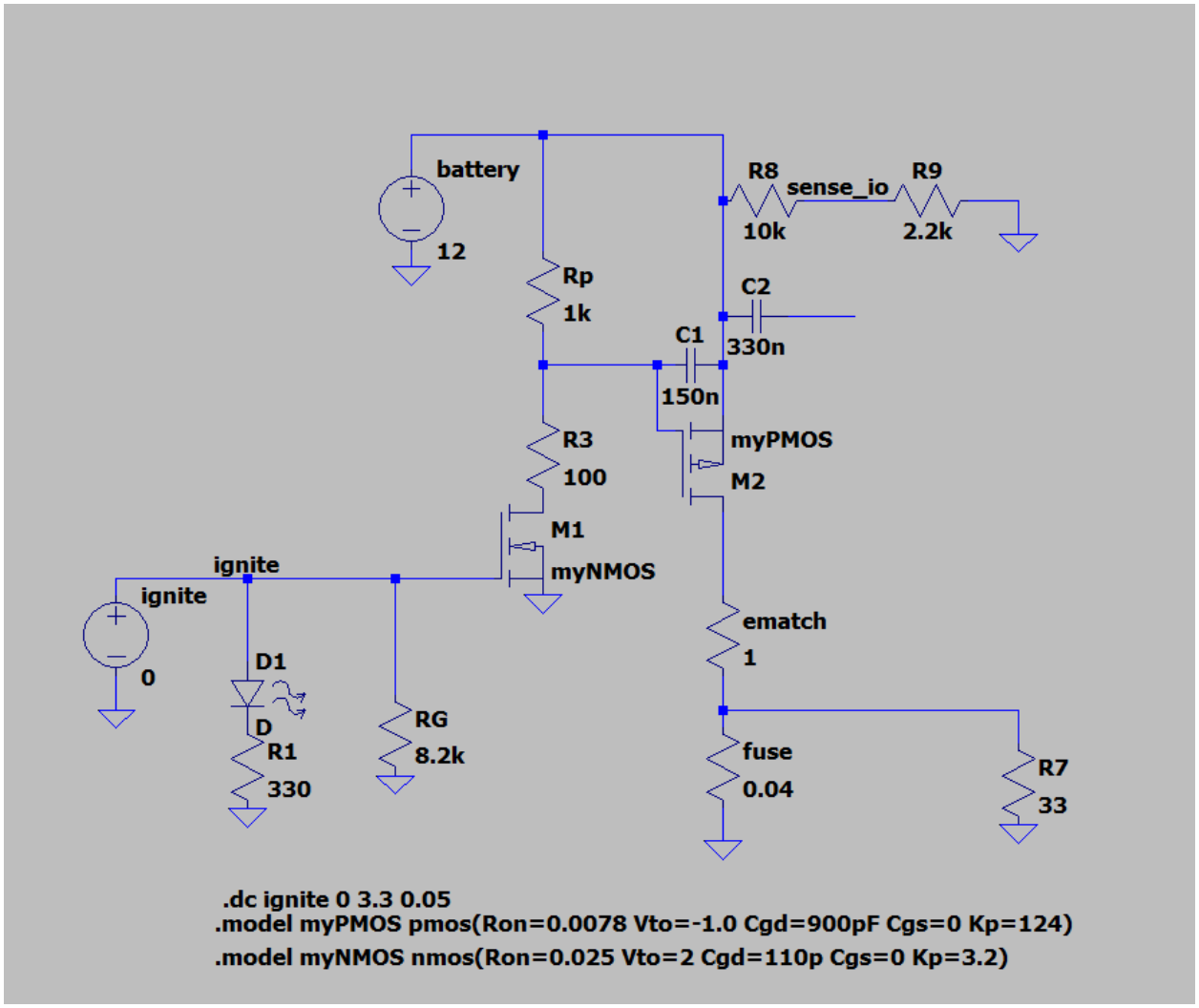
Analog Design Iterations (ts so fried)

Design 1:

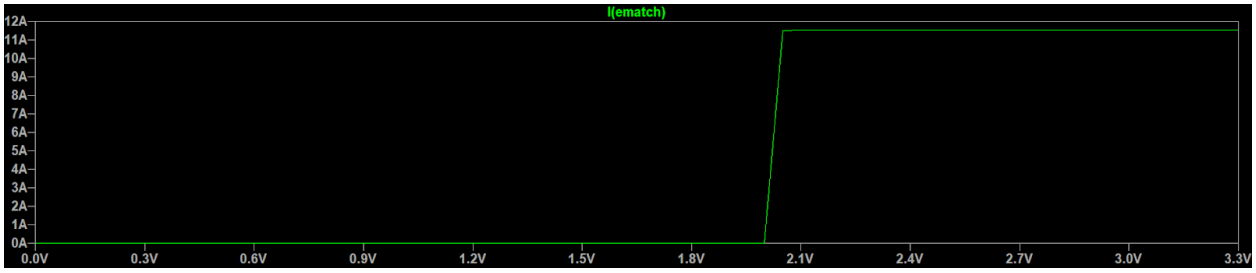
Problem: Ematch continuity sense R7 doesn't work because the current will just flow through the fuse

Features:

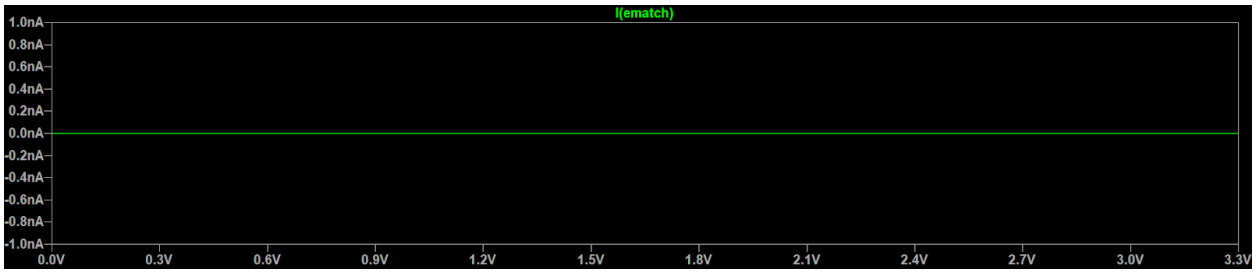
- Sensing for arming rail



Test: sweep ignite voltage with battery on: pass



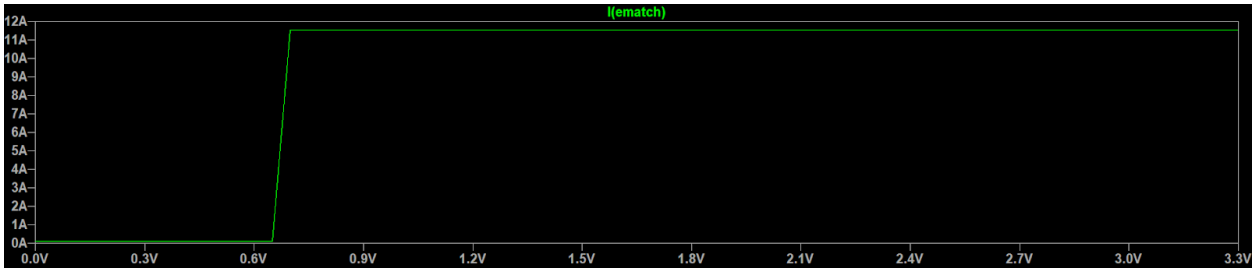
Test: test: check if ignite goes hi but battery is off, the pyro doesn't fire: pass



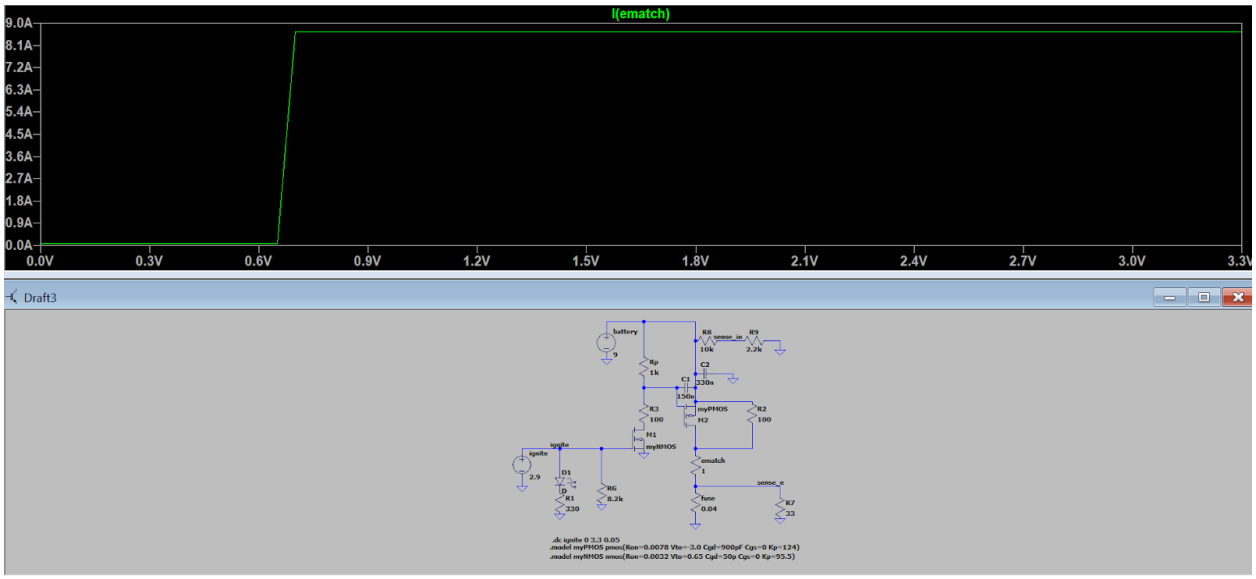
Design 2: change nmos to AO400A

Changed the nmos because the original nmos had too high of a threshold voltage compared to VOL and VOH for the mcu ignite pin

Test: sweep ignite with battery on. pass

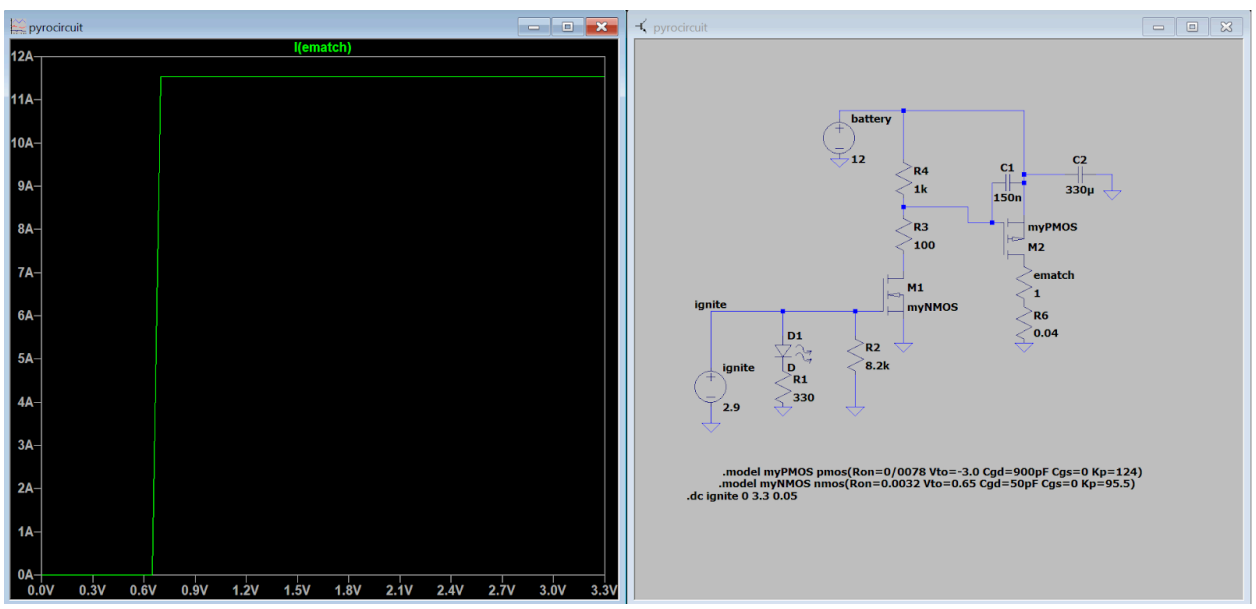


test: the pyro still fires if battery somehow sags to 9V at minimum ignite signal: pass

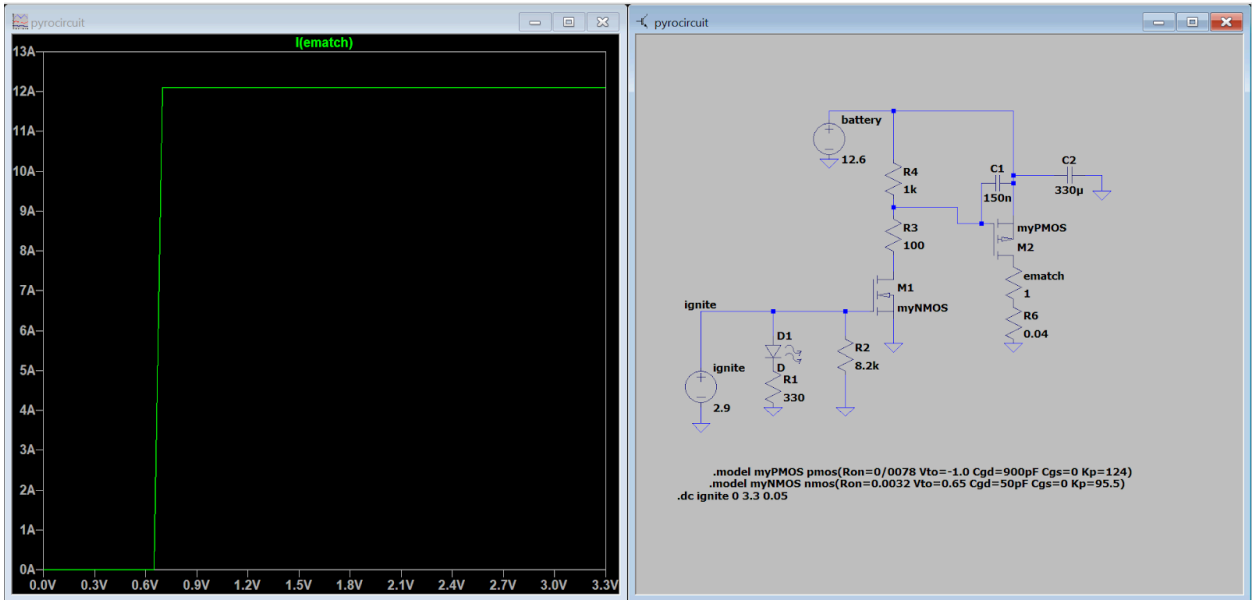


Design 3: remove R7. no ematch continuity sensing

Test: dc sweep ignite with battery on: pass

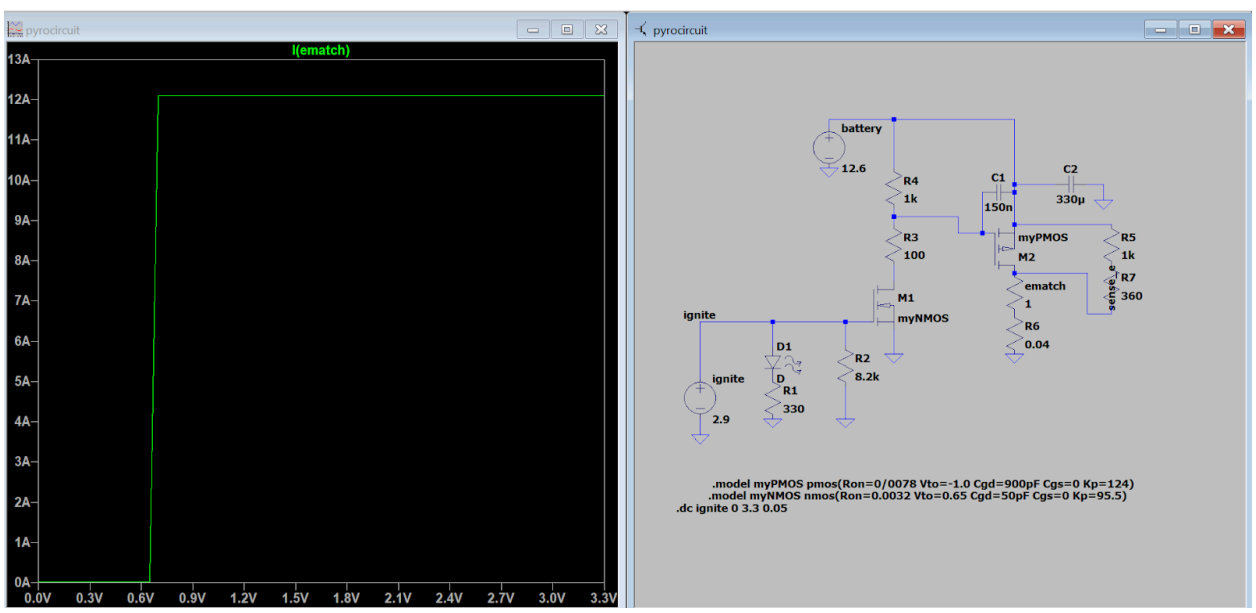


Test: dc sweep ignite at lowest thresholds and highest battery voltage: pass

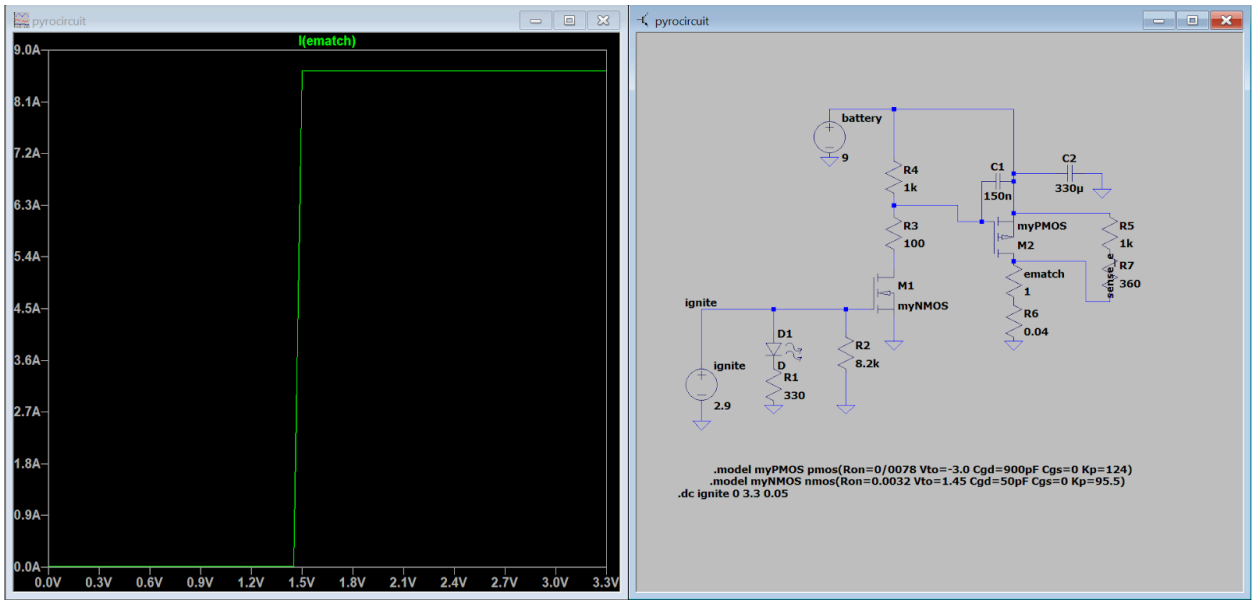


Design 4: added a sensing ematch continuity circuit. Voltage divider across the pmos.

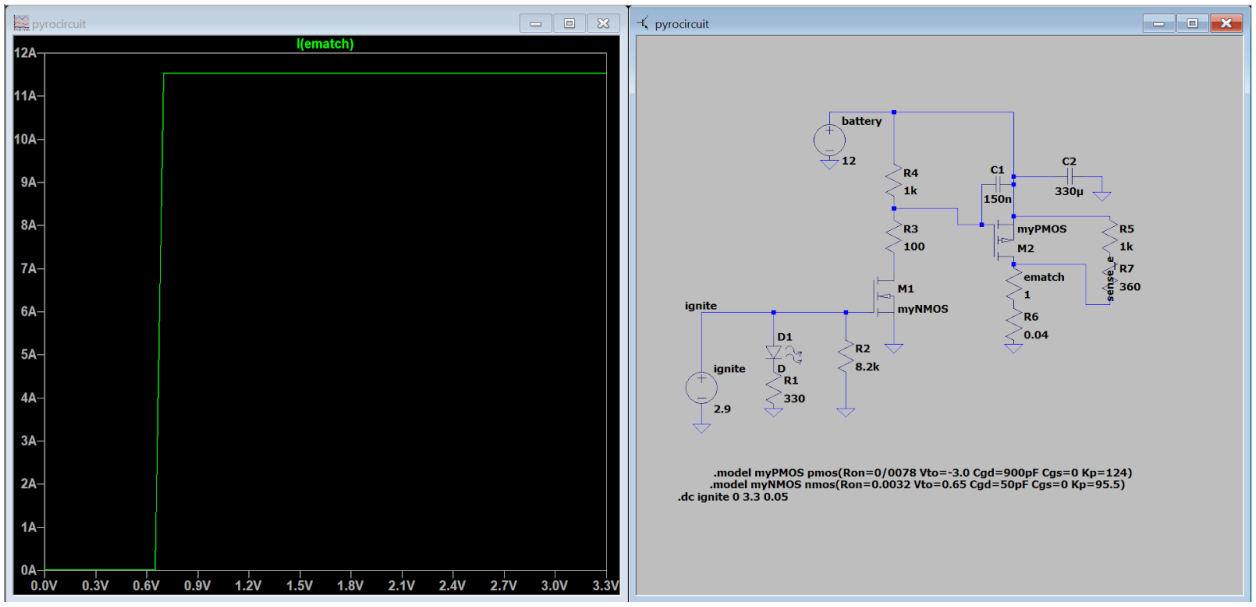
Problem: after firing the ematch, the sensing pin would go to 12V and the limit for the pin is 4V.
Test: dc sweep ignite at lowest thresholds and highest battery voltage + sense_e PASS



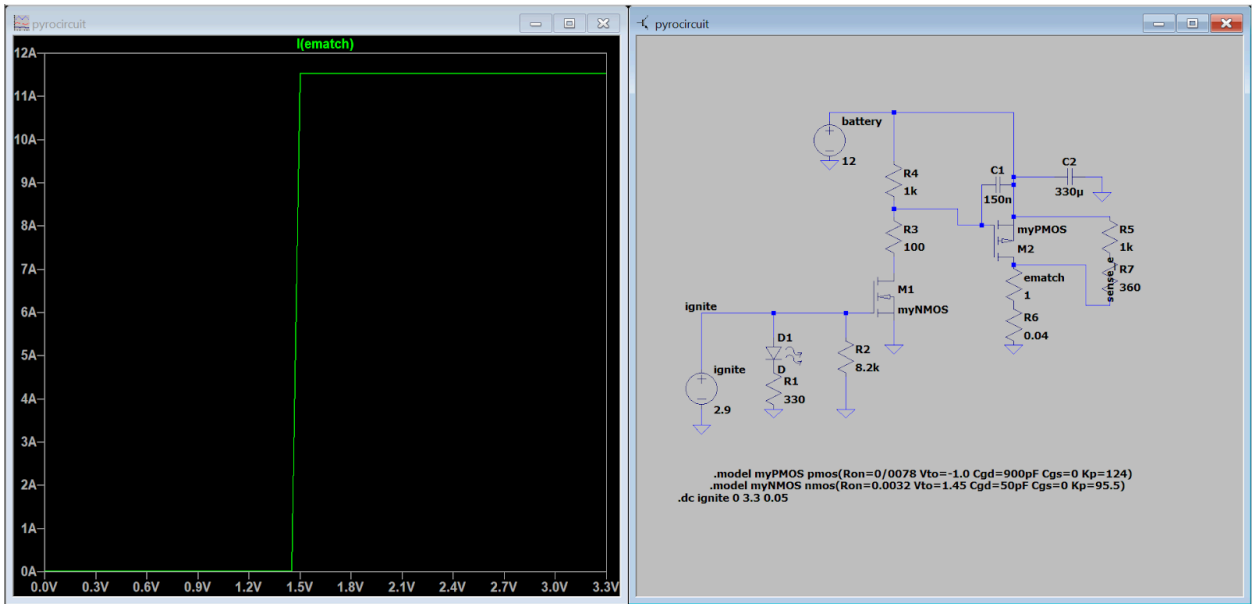
Test: dc sweep ignite highest thresholds and lowest battery voltage + sense_e : PASS



Test: nmos 0.65V, pmos -3V, battery 12V



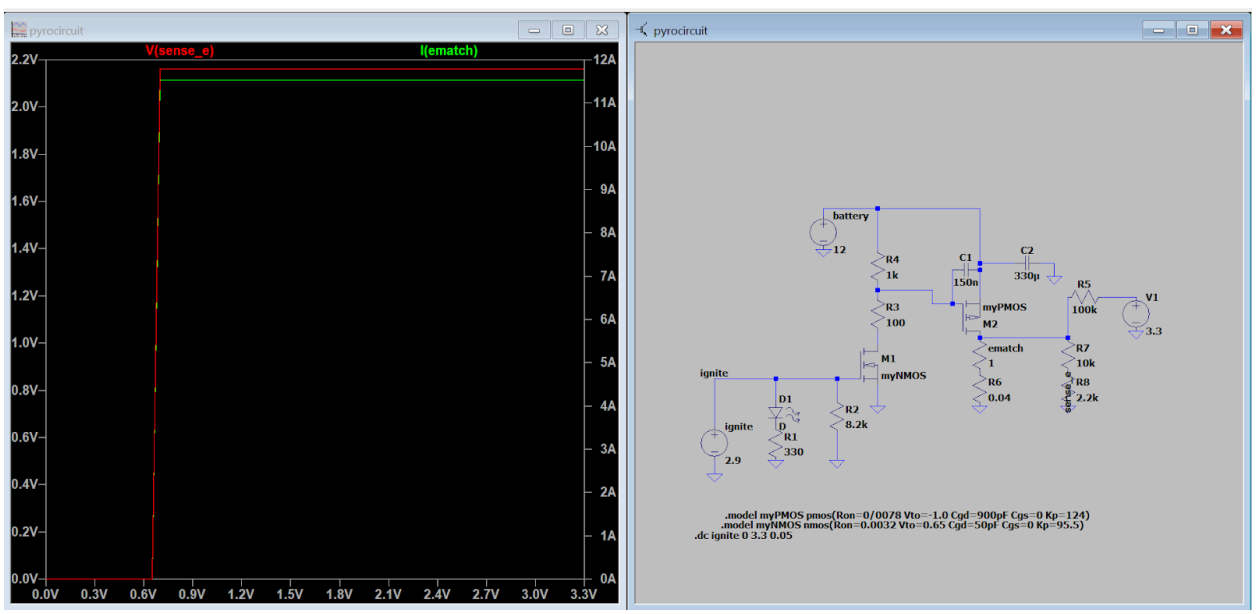
Test: nmos 1.45V, pmos 01V, battery 12V



Design 5: ematch continuity sensing using 3.3V rail and resistors

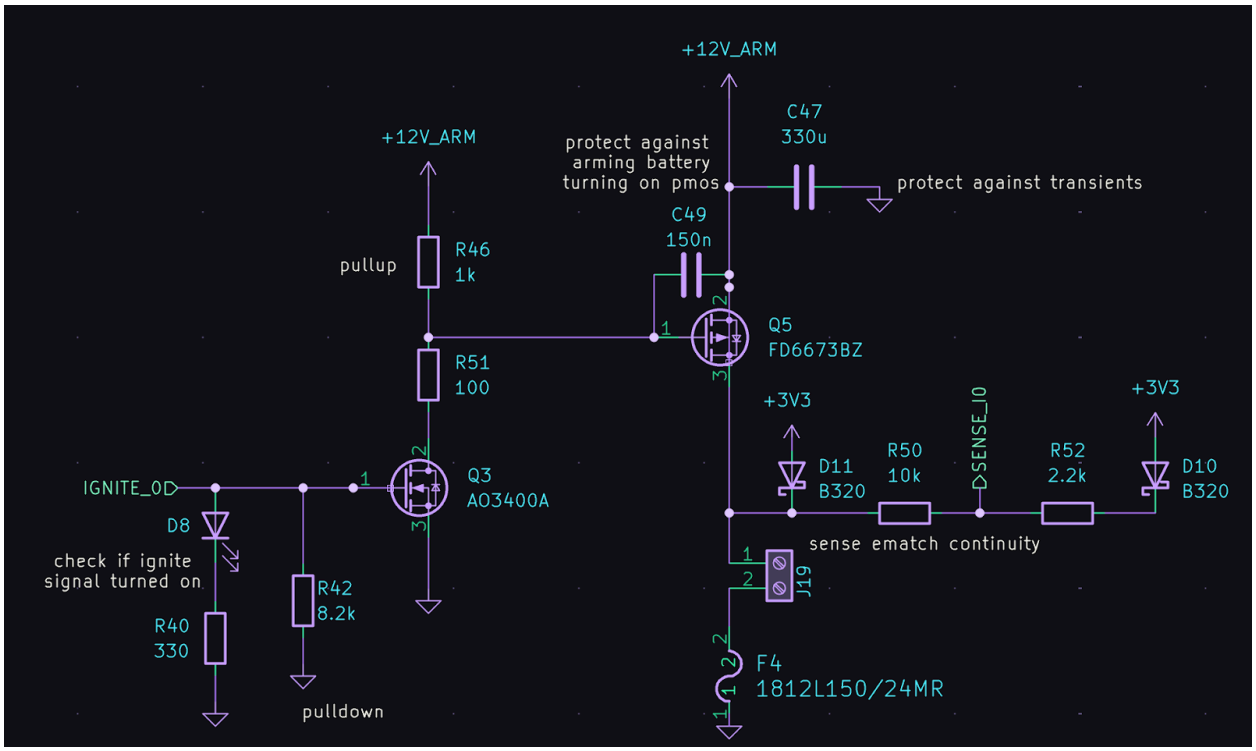
Test: dc sweep ignite: pass

Problem: the sense signal is LO both when the ematch is connected and when the ematch is not.



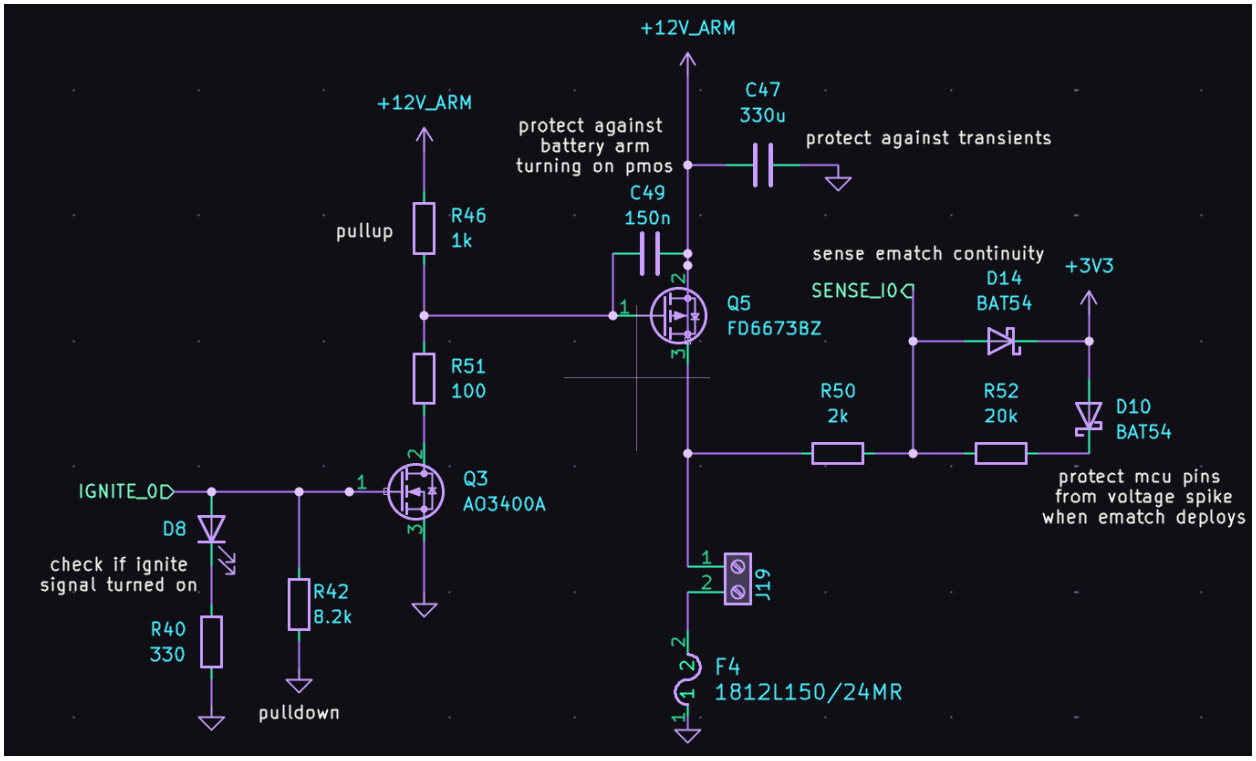
Design 6: added some diode clamps.

Problem: doesn't protect the SENSE_IO pin from 12V when firing ematch

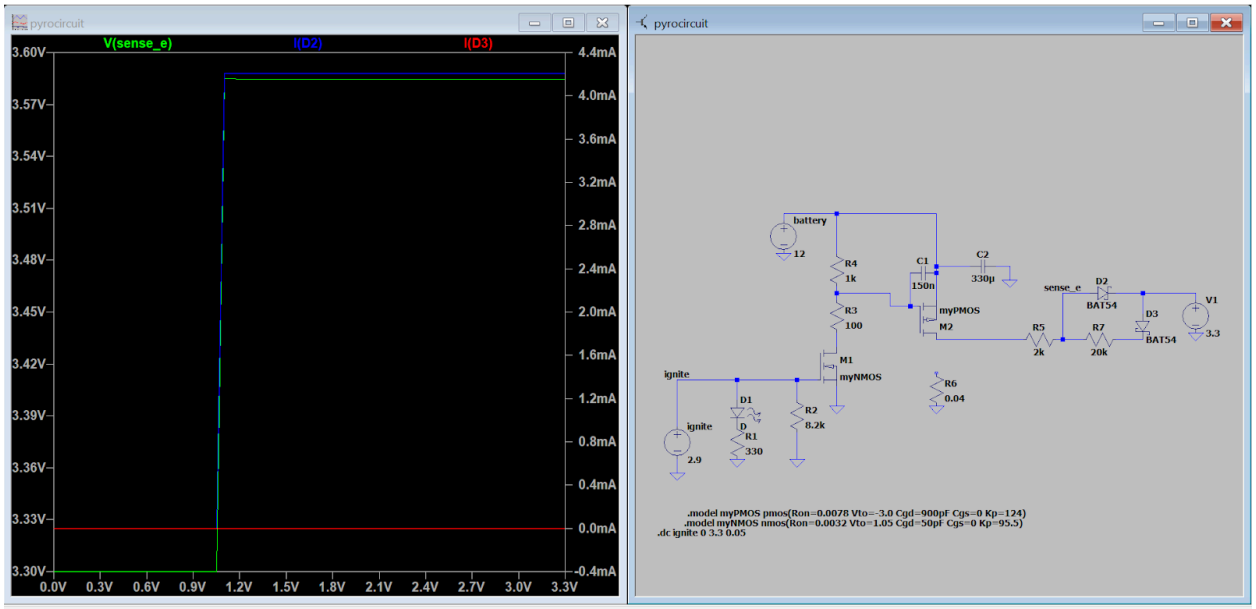


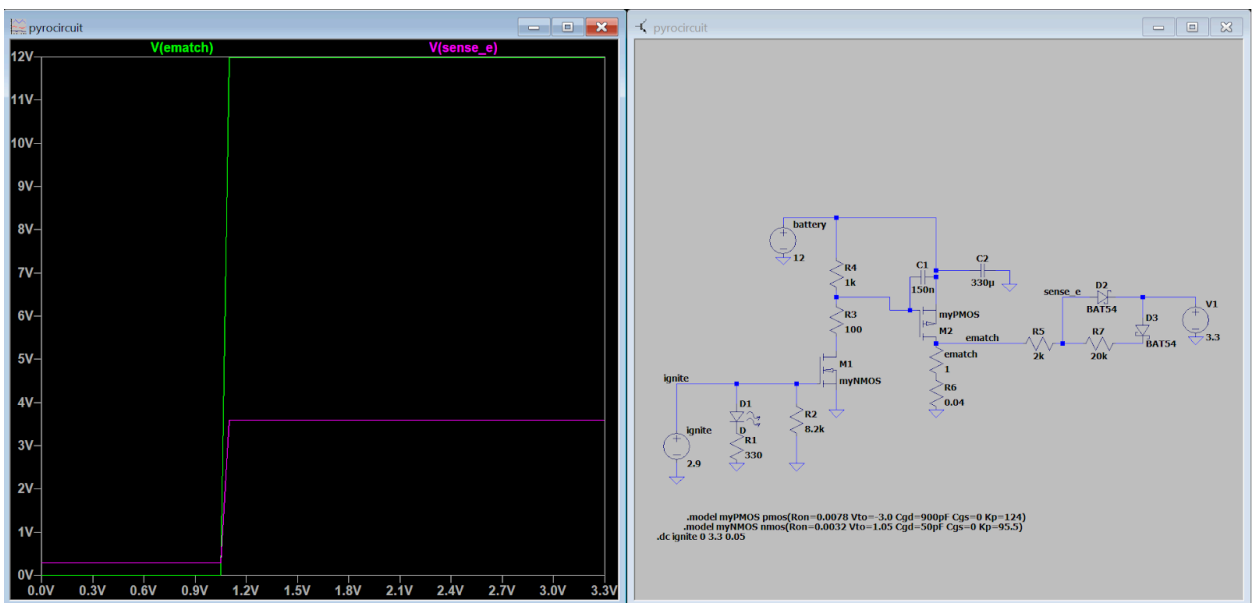
Design 7: final

```
.model myPMOS pmos(Ron=0.0078 Vto=-1.0 Cgd=900pF Cgs=0 Kp=124)
.model myNMOS nmos(Ron=0.0032 Vto=0.65 Cgd=50pF Cgs=0 Kp=95.5)
```

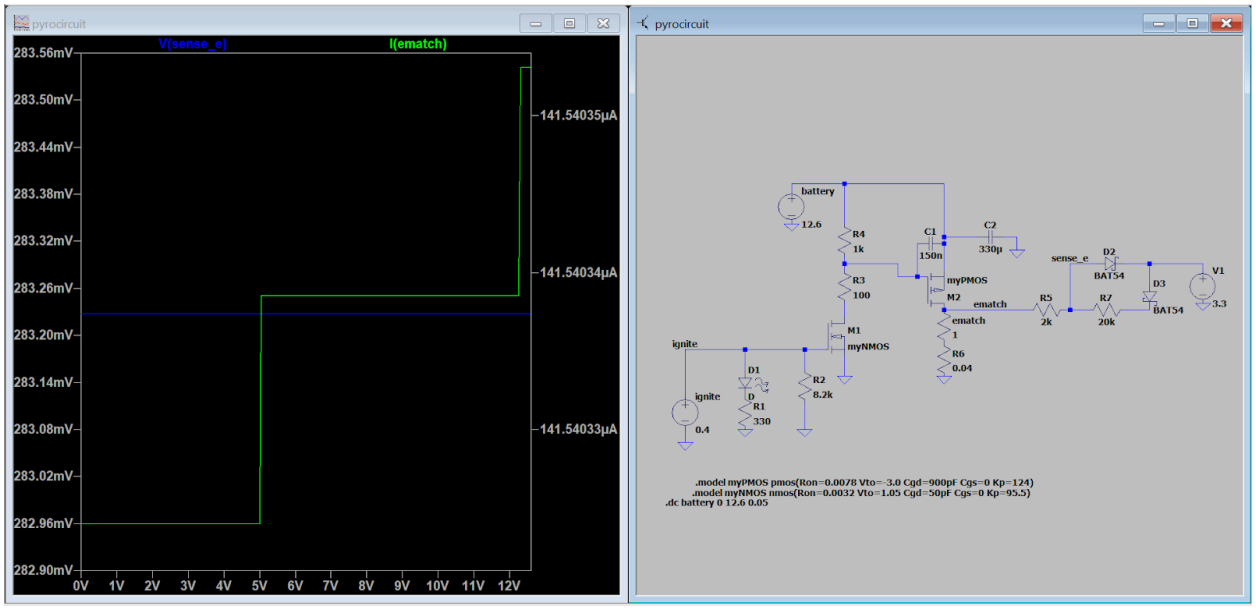


test: ematch disconnected. before and after fire. battery at 12v. PASS.

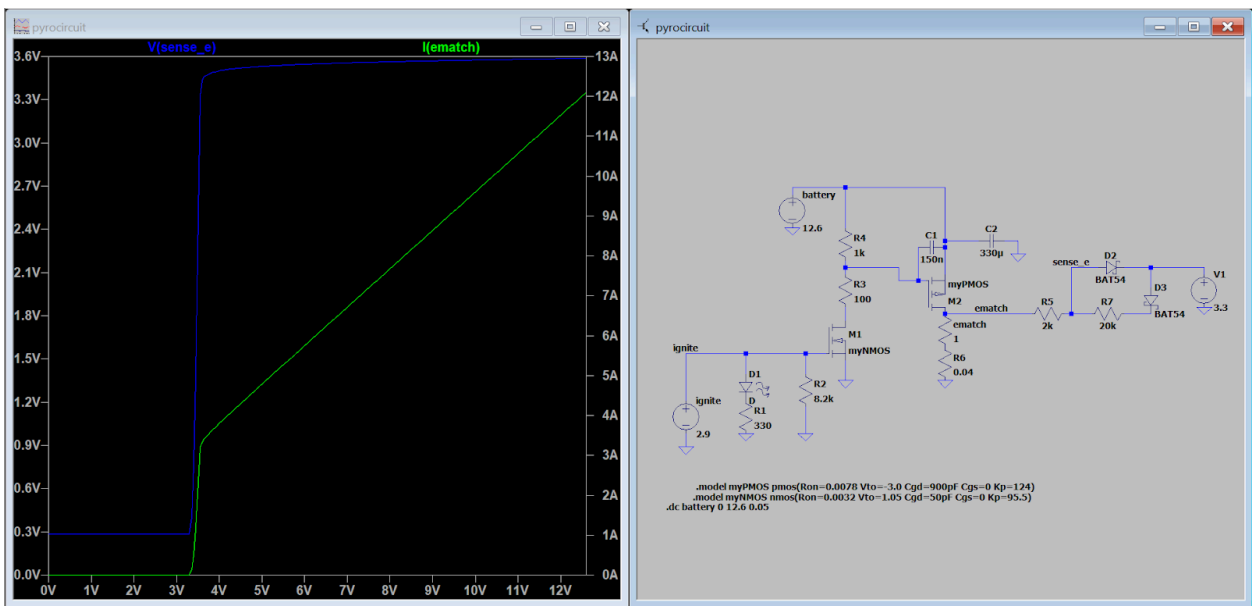




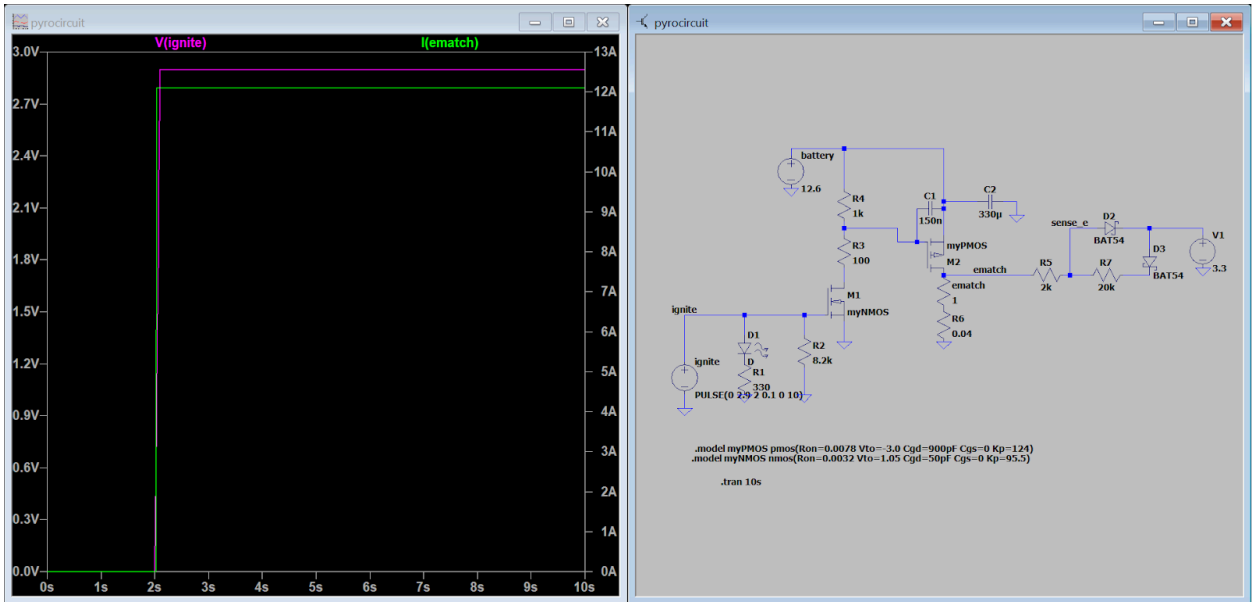
Test: ignite below 0.4V, battery on. Pass



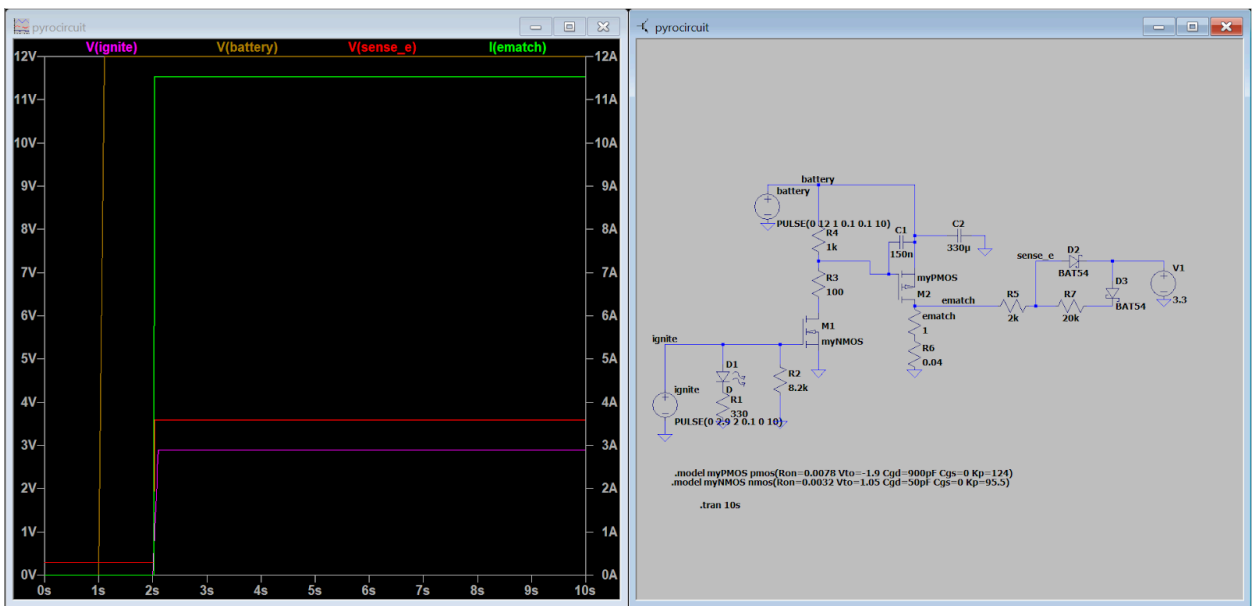
Test: ignite at 2.9V(mcu gpio pin hi threshold), battery on



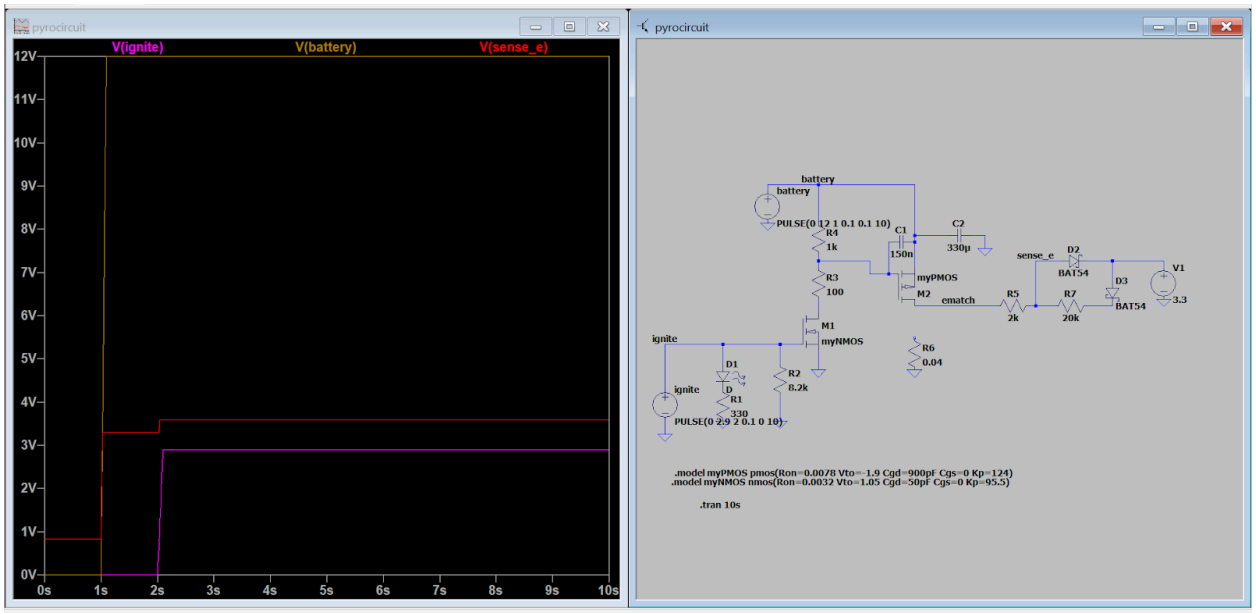
Test: inject ignite as a step function



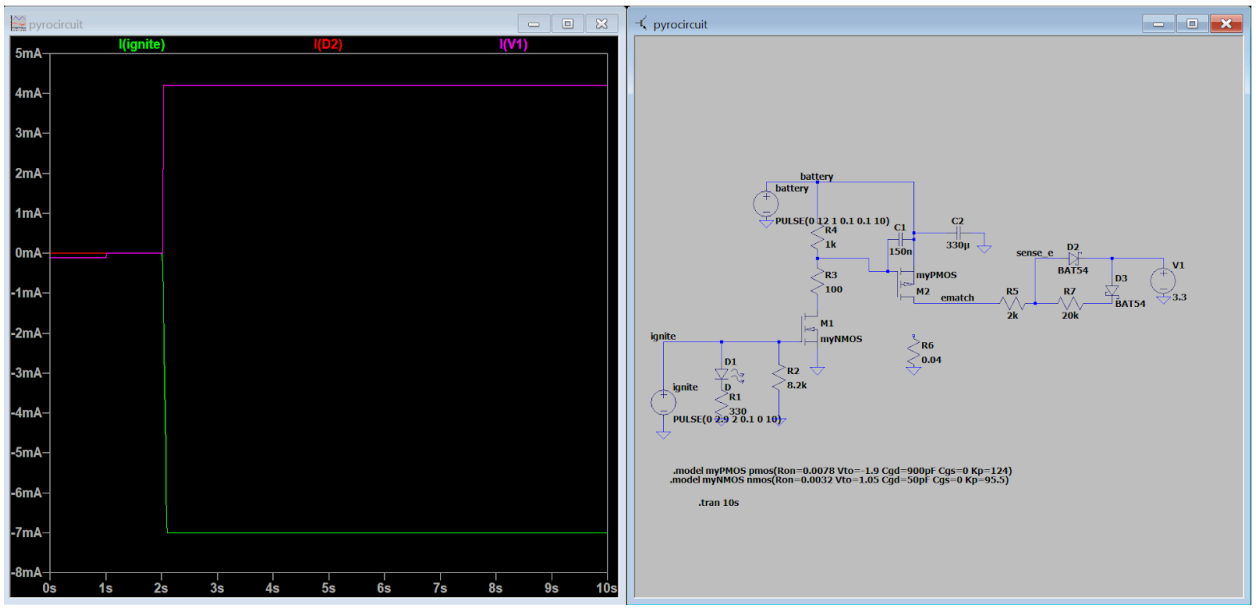
Test: arm battery (step function), turn on ignite (step function).



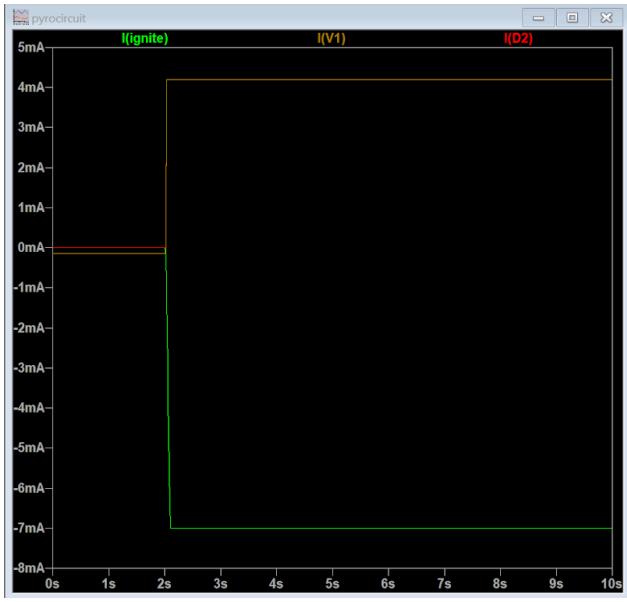
Test: ematch gone. Voltage test

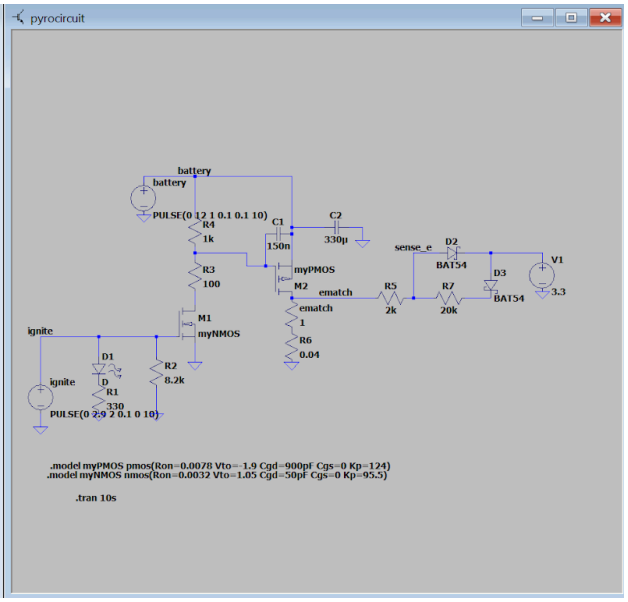


Test: ematch gone. Currents sunk/sources by mcu pins <8mA.



Test: ematch there, all thresholds average. all currents into mcu pins less than 8mA


pyrocircuit
I(ignite)
I(V1)
I(D2)
5mA
4mA
3mA
2mA
1mA
0mA
-1mA
-2mA
-3mA
-4mA
-5mA
-6mA
-7mA
-8mA
0s
1s
2s
3s
4s
5s
6s
7s
8s
9s
10s


pyrocircuit
battery
battery
PULSE(0 12 1 0.1 0.1 10)
R4
1k
C1
150n
C2
330µ
R3
100
myPMOS
M2
ematch
ematch
1
R6
0.04
M1
myNMOS
ignite
D1
D
R1
330
R2
8.2k
ignite
PULSE(0 2.9 2 0.1 0 10)
sense_e
D2
BAT54
D3
BAT54
R5
2k
R7
20k
V1
3.3
.model myPMOS pmos(Ron=0.0078 Vto=-1.9 Cgd=900pF Cgs=0 Kp=124)
.model myNMOS nmos(Ron=0.0032 Vto=1.05 Cgd=50pF Cgs=0 Kp=95.5)
.tran 10s